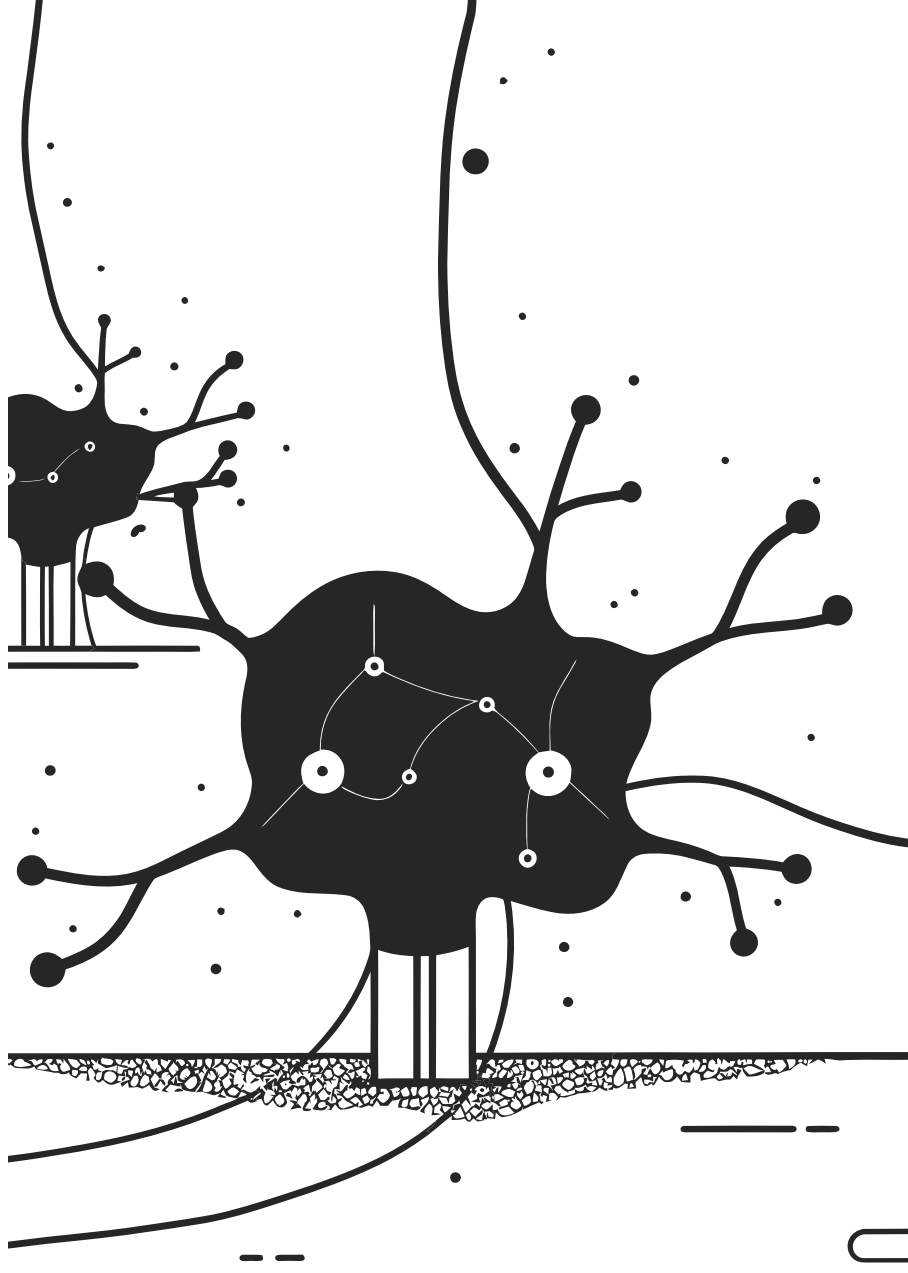
Reading the Chain: a First Block-Explorer Lab

Before you can judge any blockchain product you must be able to read what the ledger actually records - and see its limits.

EU BLOCKCHAIN OBSERVATORY 2024 | ETHERSCAN | BITCOIN EXPLORER



Core Ideas



Hash

A fixed-size fingerprint of data - every transaction has one.



Transaction

A signed instruction. The signature proves control of the address - not the person's identity.



Gas / Fee

The price of being included in a block. Paid per Ethereum transaction.



Mempool

The waiting room of unconfirmed transactions before they enter a block.



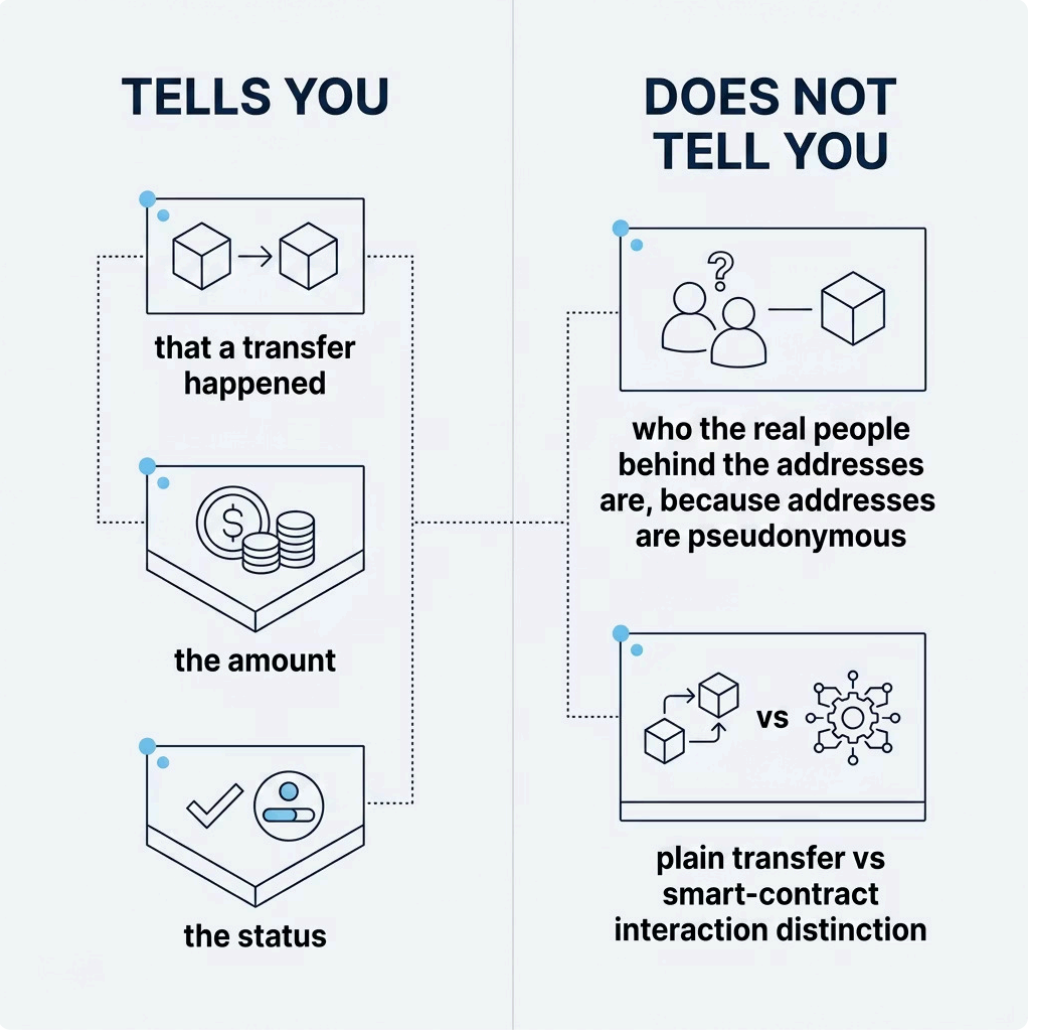
Confirmations

Later blocks added since the transaction - each one deepens settlement.

What an Explorer Shows - and Does Not

What you can see

- Sender and receiver addresses
- Value transferred
- Transaction hash (its fingerprint)
- Gas fee paid
- Confirmations - blocks added since
- Smart-contract: function called and events emitted



Worked Example

Open a transaction on Etherscan:

From

0xA1…

To

0xB2…

Value

0.5 ETH

Status

Success

Confirmations

42

You can say

The transfer succeeded and is settled.

You cannot say

Who owns 0xA1. Pseudonymity is exactly why the AML Travel Rule exists.